Circular Evaluation as Systematic Risk in Weak Supervision for Agent Progress Monitoring

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Abstract

Weak supervision pipelines for detecting progress stagnation in LLM agents face a subtle but critical evaluation pitfall: when both training labels and evaluation metrics derive from the same labeling functions (LFs), methods that reverse-engineer LF heuristics appear dominant while methods that learn genuine semantic signals are penalized. We construct a 650-window human-annotated benchmark (κ=0.76) from SWE-agent trajectories and reveal that $\sim$22% of majority-vote labels are incorrect—predominantly missing semantically diverse stagnation where agents perform varied but ineffective actions. Structural feature classifiers, which achieve near-perfect F1 (0.995) on LF-derived labels, collapse to 0.78–0.80 on gold labels (-19.5 to -21.5pp), while embedding-based probes drop only 8.6–10.1pp and *surpass* structural methods on gold (0.82–0.84 vs. 0.78–0.80). An overlap ablation isolating 12 LF-correlated features confirms that $>$55% of structural signal is circular (residual F1=0.70). Instance-isolated cross-agent evaluation reduces structural GB from 0.967 to 0.89 (-7.7pp), shifting the probe crossover point from 50% to $\sim$8% fine-tuning data. LLM-as-judge with 5-shot chain-of-thought reaches 0.81 gold F1, matching probe and structural baselines. Our results reframe the narrative from "structural dominance" to "circular evaluation as systematic risk," and contribute an annotated benchmark, a quantification methodology for circular bias, and evidence that embeddings learn beyond LF heuristics.

1 Introduction

LLM-based agents increasingly tackle complex, multi-step tasks such as software engineering **??**, web navigation, and scientific discovery. A critical operational challenge is *progress stagnation*: episodes where the agent executes plausible-looking actions that collectively fail to advance toward the goal. Unlike simple action repetition, stagnation can be *semantically diverse*—the agent tries different commands, edits files, and produces varied output, yet makes no meaningful progress.

Weak supervision (WS) offers a scalable approach to labeling stagnation at scale: domain experts encode heuristic labeling functions (LFs) such as command repetition rate and file-diff frequency, and a label model aggregates their votes **?**. In prior work on this pipeline, structural feature classifiers trained on LF-derived majority-vote (MV) labels achieved near-perfect F1 ($>$0.99) and appeared to dominate embedding-based probes and LLM-as-judge baselines by large margins.

We show that this "dominance" is largely an artifact of **circular evaluation**: structural features are derived from the same quantities that LFs compute, so a classifier trained on LF-derived labels and evaluated against those same labels is effectively solving a tautological reconstruction task. The key evidence comes from a **650-window human-annotated gold set** (κ=0.76, adjudication rate 31%), which reveals:

1. **$\sim$22% of MV labels are incorrect**, predominantly missing semantically diverse stagnation (60.1% of disagreements are MV=progress but gold=stagnation).
2. **Structural classifiers collapse on gold** (0.995$\rightarrow$0.78–0.80, -19.5 to -21.5pp), while **probes are gold-robust** (0.906$\rightarrow$0.82–0.84, -8.6 to -10.1pp) and surpass structural on gold by 2–4pp.
3. **Feature overlap ablation** confirms $>$55% of structural signal is circular (residual F1=0.70 vs. full 0.80).

4. **Instance-isolated cross-agent** evaluation drops structural GB from 0.967 to 0.89, shifting the probe crossover from 50% to $\sim$8% fine-tuning data.
5. **LLM 5-shot+CoT** reaches 0.81 gold F1, matching trained methods at $15.20 inference cost.

Our contributions are: (1) a 650-window annotated benchmark for agent stagnation detection; (2) a methodology for quantifying circular evaluation bias in WS pipelines for sequential data; (3) evidence that embedding-based probes learn genuine signals beyond LF heuristics, as revealed by their gold-robustness.

2 Related Work

Agent Progress Monitoring. Cycle detectors such as CDCS+CDSA and SpecRA capture repetitive loops but miss semantically diverse stagnation. SlopCodeBench **?** documents that coding agents exhibit structural erosion in 77% and rising verbosity in 75.5% of long trajectories, motivating window-level monitoring. Process Reward Models (PRMs) **??** train step-level evaluators via TD/GAE but require outcome labels and RL infrastructure, positioning them as training signals rather than lightweight monitoring tools.

Weak Supervision for Sequential Data. Data programming **?** and its extensions assume conditionally independent LFs. In sequential settings, LFs exhibit temporal dynamics—leading, concurrent, or lagging relationships with target events—that violate the i.i.d. assumption. Our work exposes a further challenge: when evaluation labels derive from the same LFs, *circular evaluation* systematically inflates methods whose features overlap with LF inputs.

LLM Self-Evaluation. Self-reflection approaches (Reflexion, LATS) depend on the stuck model's own judgment. External LLM-as-judge provides independent assessment but at high cost. We include LLM-as-judge as a baseline and show it converges with other methods on gold labels.

3 Problem Setup

Task. Given an agent trajectory $\tau = (a_1, o_1, \ldots, a_T, o_T)$ over a task, we segment τ into overlapping windows of $k = 5$ steps. For each window w_i, the goal is binary classification: *progress* vs. *stagnation*.

Labeling Functions. Three LFs vote on each window: (1) `cmd_rep`—command repetition ratio above threshold; (2) `file_diff`—absence of file modifications; (3) `outcome_backpropagation` (OB)—backward propagation of task outcome. Majority vote (MV) with a confidence threshold ($\geq$2 agreeing LFs) produces training labels.

Methods Under Comparison.
- **Structural classifiers**: Random Forest (RF) and Gradient Boosting (GB) over 30 hand-crafted features (command patterns, file diffs, timing, trajectory position).
- **Embedding probes**: A GRU head over frozen embeddings from Qwen3-Embedding-0.6B, E5, or BGE encoders, trained on MV labels.
- **LLM-as-judge**: Claude Sonnet 4 in zero-shot and 5-shot+CoT configurations.
- **Temporal label models** (TWV, HMM, lag-aware HMM): explored but uniformly fail to improve over MV (negative finding; see Appendix).

4 Gold Benchmark Construction

We construct a human-annotated gold set to break the circular evaluation loop.

Table 1: Gold–MV disagreement breakdown (N=143).

Type	Count	%	Dominant Pattern
MV=prog → gold=stag	86	60.1	Diverse actions, file changes, no progress
MV=stag → gold=prog	57	39.9	Repetitive but productive commands

Table 2: Method comparison on LF-derived vs. gold labels (N=650). Δpp = Gold − LF macro F1.

Method	LF F1	Gold F1	Δpp	Stag F1	Prog F1
Structural RF	.995	.78	−21.5	.68	.88
Structural GB	.995	.80	−19.5	.71	.89
Probe (Qwen)	.906	.82	−8.6	.76	.88
Probe (E5)	.936	.83	−10.6	.77	.89
Probe (BGE)	.941	.84	−10.1	.78	.90
LLM 0-shot	.663	.73	+6.8	.66	.80
LLM 5-shot+CoT	—	.81	—	.75	.87
TWV Labels	—	.72	—	.63	.81
TWV Probe	.952	.80	−15.2	.73	.87
MV Labels	1.00	.78	—	—	—

Sampling. From 15,401 SWE-agent windows with MV confidence $\geq$2, we draw a stratified sample of N=650 windows (201 stagnation, 449 progress; gold stagnation rate 30.9%).

Annotation Protocol. Two trained annotators independently label each window. Inter-annotator agreement reaches κ=0.76 (substantial agreement). 31% of windows require adjudication by a third annotator. The moderate κ reflects inherent ambiguity at stagnation boundaries—itself a finding: stagnation is not a crisp binary concept.

Gold–MV Disagreement Analysis. MV and gold agree on 507/650 windows (78%). Among the 143 disagreements (Table 1), 86 (60.1%) are MV=progress but gold=stagnation, where LFs miss *semantically diverse stagnation*: the agent performs varied commands with file changes but makes no goal-directed progress. The remaining 57 (39.9%) are MV=stagnation but gold=progress, where repetitive commands are actually productive (e.g., iterative testing). Gold stagnation rate (30.9%) exceeds MV rate (26.5%) by 4.4pp, confirming LFs systematically underestimate stagnation.

5 Experiments

All methods are re-evaluated on the 650-window gold set. We report macro F1, per-class F1, and the drop (Δpp) from LF-label to gold-label evaluation.

5.1 Gold-Validated Method Comparison

Table 2 presents the central result. On LF-derived labels, structural classifiers appear dominant (0.995 macro F1), far exceeding probes (0.91–0.94) and LLM-as-judge (0.66). On gold labels, this ranking *inverts*: Probe (BGE) leads at 0.84, followed by Probe (E5) at 0.83, Probe (Qwen) at 0.82, LLM 5-shot+CoT at 0.81, Structural GB at 0.80, and Structural RF at 0.78.

The key patterns:

- Structural classifiers suffer the largest gold drops (−19.5 to −21.5pp) because they effectively reverse-engineer LF logic: when LFs are wrong, structural classifiers follow.

Table 3: Performance on MV–gold disagreement windows (N=143).

Method	Macro F1	Follows Gold %	Follows MV %
Structural RF	.36	35.7	64.3
Structural GB	.39	38.5	61.5
Probe (Qwen)	.53	52.4	47.6
Probe (BGE)	.56	55.2	44.8
LLM Judge 0-shot	.49	48.3	51.7

Table 4: LLM judge variants vs. trained methods on gold (N=650).

Method	Gold F1	Stag F1	$\text{Recall}_{\text{stag}}$	Cost ($)
LLM 0-shot	.73	.66	.58	11.66
LLM 5-shot+CoT	.81	.75	.72	15.20
Structural GB	.80	.71	.63	0
Probe (Qwen)	.82	.76	.71	0
Probe (BGE)	.84	.78	.74	0

- Probes drop only 8.6–10.1pp and surpass structural on gold by 2–4pp, demonstrating that embeddings encode signals beyond LF heuristics.
- LLM-as-judge *improves* on gold (+6.8pp for 0-shot), confirming that its prior "errors" relative to LF labels were partially correct human-like judgments.
- All methods converge to the 0.78–0.84 range on gold, suggesting a ceiling around 0.84–0.86 imposed by annotation ambiguity (κ=0.76).

Disagreement Window Analysis. On the 143 MV-gold disagreement windows (Table 3), structural classifiers follow MV errors (macro F1 0.36–0.39, following MV $>$60%), while probes align more with gold (0.53–0.56, following gold $>$52%). Probe "false positives" on LF labels—cases where the probe calls stagnation but LFs say progress—are validated by gold 42% of the time: the probe detected real stagnation that LFs missed.

5.2 LLM-as-Judge: Few-Shot Chain-of-Thought

We evaluate Claude Sonnet 4 with 5 in-context examples (2 stagnation, 2 progress, 1 ambiguous boundary case) and chain-of-thought prompting. Table 4 compares LLM variants with trained methods on gold.

Five in-context examples teach the LLM the concept of "semantically diverse stagnation"—the exact pattern that LFs miss. CoT prompting raises stagnation recall from 0.58 to 0.72 by forcing explicit reasoning about goal-directed progress. At 0.81 gold F1, the few-shot LLM matches Structural GB (0.80) and falls within 1–3pp of probes.

Rank Inversion. On LF labels, structural GB leads LLM by $\sim$33pp (0.995 vs. $\sim$0.66). On gold, the LLM *surpasses* structural GB by 1pp (0.81 vs. 0.80). Three fundamentally different approaches—hand-crafted features + ML, frozen embeddings + GRU, and in-context learning—converge to 0.80–0.84 on gold, suggesting this range reflects the task's achievable ceiling given annotation ambiguity.

5.3 Structural Feature Overlap Ablation

To quantify circular evaluation, we partition the 30 structural features into 12 *overlap* features (directly correlated with LF inputs: command repetition ratios, file diff counts, etc.) and 18 *residual* features

Table 5: Overlap ablation: LF-label vs. gold-label performance (N=650).

Feature Set	N_f	**LF F1**	**Gold F1**	**LF–Gold Gap**
Full structural	30	.995	.80	.195
Overlap-only	12	.98	.77	.21
Residual (non-overlap)	18	.87	.70	.17
Probe (Qwen, ref.)	—	.906	.82	.086
Probe (BGE, ref.)	—	.941	.84	.101

Table 6: Top-5 residual features by gold-label importance.

Feature	**Gini**	**Gold Univ. F1**	r **w/ cmd_rep**
action_diversity_entropy	.18	.62	.61
trajectory_position_norm	.14	.59	.08
mean_output_length	.12	.57	.23
step_duration_variance	.09	.56	.15
action_length_mean	.08	.55	.31

(action diversity, trajectory position, output length, timing). Table 5 reports performance under each feature set.

Removing the 12 overlap features drops gold F1 from 0.80 to 0.70 ($-$10pp), confirming that $>$55% of structural signal on gold passes through LF-correlated channels. The residual 0.70 exceeds random (0.50) by 20pp—structural features are not vacuous—but falls 12–14pp below probes, demonstrating that hand-crafted non-circular features cannot compete with learned semantic representations.

The overlap-only features achieve 0.98 on LF labels but only 0.77 on gold (gap = 0.21), the largest LF–gold gap of any configuration—a direct measure of circular inflation. Among residual features, action diversity entropy (Gini importance 0.18, gold univariate F1 = 0.62) and trajectory position (0.14, 0.59) carry the most genuine signal.

5.4 Cross-Agent Instance-Isolated Evaluation

Prior cross-agent evaluation (training on SWE-agent, evaluating on OpenHands) did not control for shared SWE-bench task instances. We find that 847 instances (28.3%) overlap between the training and evaluation sets. These shared instances are easier (mean solve rate 42% vs. 28%) and have lower stagnation rates (19% vs. 23%), inflating cross-agent scores for methods that exploit task-level surface similarity.

Table 7 reports results after removing all shared-instance windows from the evaluation set.

Structural GB drops 7.7pp to 0.89; probes are minimally affected ($\leq$1.8pp). The probe crossover point shifts dramatically: in the original evaluation, 50% fine-tuning data was needed to match structural GB; under instance isolation, **10% fine-tuning (0.91) already surpasses structural GB (0.89)** by 2pp.

Structural features still dominate zero-shot transfer (+31pp over probe 0-shot), confirming their value when no target-agent data is available. However, the "50% FT to catch up" narrative was an artifact of task overlap: the true crossover is $\sim$8% FT, requiring only tens of labeled windows in practice.

6 Discussion

Circular Evaluation as Systematic Risk. Our results establish a three-pronged evidence chain against evaluating WS-trained classifiers on LF-derived labels for sequential data: (i) gold validation shows 22% label error concentrated on semantically diverse stagnation; (ii) overlap ablation quantifies $>$55% of structural signal as circular; (iii) instance isolation reveals 7.7pp task-overlap inflation. Together, these

Table 7: Instance isolation impact on cross-agent transfer (MV labels).

Method	Original	Isolated	Δpp
Structural GB	.967	.89	-7.7
Structural RF	.912	.86	-5.2
Probe 0-shot	.598	.58	-1.8
Probe 5% FT	.850	.84	-1.0
Probe 10% FT	.918	.91	-0.8
Probe 20% FT	.948	.94	-0.8
Probe 50% FT	.968	.96	-0.8

Table 8: Crossover analysis: Structural GB vs. Probe under isolation.

Comparison	Orig. Gap	Isol. Gap	Shift
GB vs. Probe 0-shot	+36.9	+31.0	narrowed 5.9pp
GB vs. Probe 5% FT	+11.7	+5.0	narrowed 6.7pp
GB vs. Probe 10% FT	+5.0	-2.0	**reversed**
GB vs. Probe 20% FT	+2.0	-5.0	reversed, widens
GB vs. Probe 50% FT	-0.1	-7.0	probe widens

artifacts transformed "probe modestly outperforms structural" (on gold: 0.82–0.84 vs. 0.78–0.80) into "structural dominates by 6–9pp" (on LF labels: 0.995 vs. 0.906–0.941).

Probe Gold-Robustness. Probes trained on noisy MV labels retain signals beyond LF heuristics. Of probe "false positives" (predicting stagnation where LFs say progress), 42% are validated by gold as genuine stagnation. This suggests that pre-trained embeddings encode goal-directedness information that survives noisy supervision—a positive finding for WS practitioners, as it means downstream models can partially self-correct LF errors.

Method Convergence at the Annotation Ceiling. All methods converge to 0.78–0.84 on gold, likely bounded by annotation ambiguity (κ=0.76, adjudication rate 31%). This convergence suggests diminishing returns from method innovation without improving label quality or task formulation.

Practical Deployment Recommendations. Based on the complete evidence:
- **Zero target-agent data**: Structural features remain the best option (0.89 isolated cross-agent F1, far above probe 0-shot at 0.58).
- **$\geq$10% target data**: Probe surpasses structural (0.91 vs. 0.89) and improves with more data.
- **Evaluation**: Gold-standard annotation is essential; LF-derived metrics overestimate structural methods by 19–21pp.

7 Conclusion

We presented a systematic investigation of circular evaluation bias in weak supervision for agent progress monitoring. Through a 650-window human-annotated benchmark, feature overlap ablation, and instance-isolated cross-agent evaluation, we demonstrated that the apparent dominance of structural features is largely an artifact of evaluating on LF-derived labels. Embedding-based probes exhibit surprising gold-robustness, surpassing structural classifiers by 2–4pp on human-annotated labels despite being trained on the same noisy supervision. Our work highlights circular evaluation as a systematic risk in WS pipelines for sequential data and provides both a benchmark and quantification methodology for future research.

Limitations. The gold set covers a single agent (SWE-agent) on software engineering tasks; generalization to other domains remains to be established. Cross-agent evaluation involves only two agents (SWE-agent and OpenHands). κ=0.76 reflects genuine task ambiguity but limits the achievable evaluation ceiling. The temporal label model negative finding is conditioned on a 3-LF, low-conflict regime and may not generalize to richer LF sets.